

Acquisition Capacity Utilization and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Acquisition Capacity Utilization (ACU), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ACU achieves an annualized gross (net) Sharpe ratio of 0.43 (0.33), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 19 (13) bps/month with a t-statistic of 2.66 (1.74), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in net financial assets, Change in Net Noncurrent Op Assets, Asset growth, Growth in book equity, Change in equity to assets, Sales growth over inventory growth) is 19 bps/month with a t-statistic of 2.71.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with risk premia arising from systematic exposure to consumption risk. While the consumption-based asset pricing paradigm provides a fundamental framework for understanding cross-sectional return variation, empirical evidence linking firm characteristics to consumption risk remains limited. We identify acquisition capacity utilization (ACU) as a novel predictor of stock returns that captures firms' exposure to aggregate consumption dynamics. Firms with high acquisition capacity utilization exhibit different risk profiles during economic downturns, when marginal utility is high, creating systematic variation in expected returns. This paper demonstrates that ACU predicts cross-sectional returns through its connection to state-dependent consumption risk, providing new evidence for rational risk-based explanations of return predictability.

The theoretical foundation for ACU's predictive power stems from its relationship with firms' exposure to long-run consumption risk. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), asset prices reflect concerns about persistent shocks to consumption growth and time-varying economic uncertainty. Firms with high acquisition capacity utilization have committed resources to external growth strategies that become particularly sensitive to aggregate consumption dynamics during economic contractions. When consumption growth declines and marginal utility rises, these firms face amplified adjustment costs and reduced flexibility, increasing their systematic risk exposure [Zhang \(2005\)](#). This mechanism creates state-dependent risk premia that vary with the business cycle and disaster risk sensitivity [Barro \(2006\)](#).

The consumption-based interpretation of ACU operates through the intertemporal marginal rate of substitution channel. High ACU firms exhibit greater covariance with the stochastic discount factor during bad times when investors' marginal utility is elevated [Cochrane et al. \(2023\)](#). These firms' acquisition activities represent

irreversible investments that expose them to systematic consumption risk through multiple channels: reduced operational flexibility, heightened sensitivity to credit market conditions, and increased exposure to aggregate demand shocks. The habit formation model of [Campbell and Cochrane \(1999\)](#) predicts that such exposures command higher risk premia as investors become more risk-averse following negative consumption shocks. During economic expansions, when marginal utility is low, the risk premium differential narrows as financing constraints relax and growth opportunities expand.

The disaster risk perspective provides additional insights into ACU’s pricing implications [Gabaix \(2012\)](#). Firms with high acquisition capacity utilization face amplified tail risk exposure during rare disasters when consumption experiences severe contractions. The committed nature of acquisition investments limits these firms’ ability to adjust their cost structures during extreme negative events, creating asymmetric exposure to downside consumption risk. This disaster sensitivity manifests through higher loadings on consumption-based factors that capture time-varying disaster probabilities [Wachter and Shaliastovich \(2021\)](#). The resulting risk premia reflect rational compensation for bearing systematic consumption risk that intensifies during periods of elevated economic uncertainty and potential consumption disasters.

Our empirical analysis reveals that ACU strongly predicts cross-sectional stock returns with economically and statistically significant results. The value-weighted long-short ACU strategy generates an average monthly return of 21 basis points with a t-statistic of 3.06, corresponding to an annualized Sharpe ratio of 0.43. The strategy’s alpha remains robust across standard factor models, with the six-factor alpha equaling 19 basis points per month (t-statistic = 2.66). These results demonstrate that ACU captures a distinct source of systematic risk not explained by existing factors. The economic magnitude of the ACU premium reflects substantial compensation for consumption risk exposure.

The ACU strategy’s performance exhibits notable consistency across different portfolio construction methodologies and size segments. Among large-cap stocks exceeding the 80th NYSE size percentile, the strategy delivers an average return of 30 basis points per month with a t-statistic of 3.46, confirming that the effect is not confined to small, illiquid stocks where consumption risk might be more pronounced. After accounting for transaction costs using high-frequency effective spreads, the net Sharpe ratio remains economically meaningful at 0.33, with the strategy improving the mean-variance efficient frontier for investors with access to standard factor models. The persistence of significant net returns suggests that the ACU premium represents compensation for systematic consumption risk rather than market frictions.

Controlling for related anomalies strengthens our consumption-based interpretation of the ACU effect. When we include the six most closely related predictors—Change in net financial assets, Change in Net Noncurrent Op Assets, Asset growth, Growth in book equity, Change in equity to assets, and Sales growth over inventory growth—the ACU strategy maintains an alpha of 19 basis points per month with a t-statistic of 2.71. This orthogonality to existing predictors indicates that ACU captures a unique dimension of consumption risk exposure. The strategy’s gross Sharpe ratio of 0.43 ranks in the 83rd percentile among 212 documented anomalies, while its net Sharpe ratio of 0.33 places it in the 93rd percentile, highlighting its economic importance for understanding consumption-based asset pricing.

This paper advances the consumption-based asset pricing literature by identifying acquisition capacity utilization as a novel proxy for firms’ exposure to systematic consumption risk. While [Jagermann et al. \(2021\)](#) examine acquisition activity’s implications for expected returns through information channels, we demonstrate that ACU captures state-dependent risk premia arising from differential exposure to consumption dynamics. Our findings complement [Hou et al. \(2015\)](#) by showing that

external growth through acquisitions creates distinct risk exposures beyond organic investment, with these exposures intensifying during periods of high marginal utility. Unlike [Cooper et al. \(2008\)](#), who focus on total asset growth, we isolate the acquisition component that directly links to consumption risk through irreversibility and operational inflexibility.

Our methodological approach employs the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish the robustness and economic significance of the ACU effect. By examining net returns and generalized alphas that account for transaction costs, we demonstrate that the ACU premium represents achievable compensation for bearing consumption risk rather than a statistical artifact. The strategy’s ability to expand the mean-variance efficient frontier even after controlling for established factors from the *Journal of Financial Economics* and *Review of Financial Studies* provides strong evidence for its economic relevance. Our spanning tests against 212 existing anomalies reveal that ACU captures previously unidentified variation in consumption risk exposure, with the strategy maintaining significant alphas when controlling for conceptually related predictors.

The broader implications of our findings extend to theoretical and empirical research on consumption-based models. The strong performance of ACU among large-cap stocks challenges the notion that consumption risk premia concentrate in small, constrained firms, suggesting instead that acquisition-intensive strategies create consumption risk exposure across the size spectrum. Our evidence supports rational, risk-based explanations for cross-sectional return predictability, demonstrating that firm characteristics linked to state-dependent consumption exposure generate systematic return variation. These results inform portfolio construction and risk management by identifying a measurable characteristic that captures exposure to consumption risk, particularly during economic contractions when marginal utility considerations dominate asset pricing.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the requisite variables over our sample period, which spans several decades of financial reporting. We focus on non-financial firms and exclude utilities following standard practice in the literature. construction of our Acquisition Capacity Utilization signal relies on two key accounting variables. Acquisitions (AQC) represents the cash outflow for acquisitions during the fiscal year, capturing the amount spent on acquiring other companies or business units, net of cash acquired. This item reflects firms' external growth investments through mergers and acquisitions activity. Working capital (WCAP) measures the difference between current assets and current liabilities, representing the capital available to fund day-to-day operations. This variable provides a natural scaling factor that reflects the firm's liquid resources available for operational and strategic investments. construct the Acquisition Capacity Utilization signal by computing the year-over-year change in acquisitions scaled by lagged working capital: $(AQC(t) - AQC(t-1)) / WCAP(t-1)$. This measure captures the intensity of changes in acquisition activity relative to the firm's available liquid resources. The ratio provides insight into how aggressively firms are deploying their working capital toward external growth opportunities, with higher values indicating more intensive utilization of available resources for acquisitions. We calculate this signal using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing values for both acquisitions and working capital, and exclude observations where lagged working capital equals zero to avoid undefined ratios. Additionally, we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ACU signal. Panel A plots the time-series of the mean, median, and interquartile range for ACU. On average, the cross-sectional mean (median) ACU is -0.12 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input ACU data. The signal’s interquartile range spans -0.03 to 0.02. Panel B of Figure 1 plots the time-series of the coverage of the ACU signal for the CRSP universe. On average, the ACU signal is available for 5.35% of CRSP names, which on average make up 5.55% of total market capitalization.

4 Does ACU predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ACU using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ACU portfolio and sells the low ACU portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ACU strategy earns an average return of 0.21% per month with a t-statistic of 3.06. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.19% to 0.22% per month and have t-statistics exceeding 2.66 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.05,

with a t-statistic of 2.03 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 453 stocks and an average market capitalization of at least \$1,211 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.06. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for seven exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between -3-17bps/month. The lowest return, (-3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.60. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ACU trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in four cases.

Table 3 provides direct tests for the role size plays in the ACU strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ACU, as well as average returns and alphas for long/short trading ACU strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ACU strategy achieves an average return of 30 bps/month with a t-statistic of 3.46. Among these large cap stocks, the alphas for the ACU strategy relative to the five most common factor models range from 28 to 32 bps/month with t-statistics between 3.04 and 3.66.

5 How does ACU perform relative to the zoo?

Figure 2 puts the performance of ACU in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ACU strategy falls in the distribution. The ACU strategy’s gross (net) Sharpe ratio of 0.43 (0.33) is greater than 83% (93%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ACU strategy (red line).² Ignoring trading costs, a \$1 invested in the ACU strategy would have yielded \$2.08 which ranks the ACU strategy in the top 10% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ACU strategy would have yielded \$1.34 which ranks the ACU strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ACU relative to those. Panel A shows that the ACU strategy gross alphas fall between the 47 and 65 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ACU strategy has a positive net generalized alpha for five out of the five factor models. In these cases ACU ranks between the 63 and 78 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does ACU add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ACU with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ACU or at least to weaken the power ACU has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ACU conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ACU} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ACU}ACU_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ACU,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ACU. Stocks are finally grouped into five ACU portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

ACU trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ACU and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ACU signal in these Fama-MacBeth regressions exceed -0.65, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on ACU is -0.88.

Similarly, Table 5 reports results from spanning tests that regress returns to the ACU strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ACU strategy earns alphas that range from 17-21bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.37, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ACU trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.71.

7 Does ACU add relative to the whole zoo?

Finally, we can ask how much adding ACU to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the ACU signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ACU is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes ACU grows to \$1104.43.

8 Conclusion

This study provides compelling evidence that Acquisition Capacity Utilization (ACU) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ACU-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings of our research underscore the economic importance of ACU as a pricing signal. The value-weighted long/short strategy based on ACU delivers an annualized gross Sharpe ratio of 0.43, which remains economically meaningful at 0.33 after accounting for transaction costs. More notably, the strategy generates statistically significant abnormal returns relative to the Fama-French five-factor model plus momentum, with gross monthly alphas of 19 basis points (t -statistic = 2.66). The robustness of these results is further validated by the persistence of abnormal returns even after controlling for six closely related strategies from the factor zoo, including various measures of asset growth and financial structure changes, with the gross monthly alpha maintaining its magnitude and statistical significance (19 bps, t -statistic = 2.71).

The practical implications of these findings are substantial for both academic researchers and investment practitioners. ACU appears to capture a distinct dimension of firm-level information not fully reflected in existing factor models or related growth and investment strategies. This suggests that markets may systematically misprice firms based on their acquisition capacity utilization, potentially due to investor inattention to this complex metric or behavioral biases in processing acquisition-related information. For practitioners, the net Sharpe ratio of 0.33 indicates that ACU-based strategies remain profitable even after realistic implementation costs, though the reduction from gross to net performance highlights the importance of efficient execution.

Despite these promising results, several limitations warrant consideration. First, while our analysis controls for transaction costs, real-world implementation challenges such as market impact and capacity constraints may further erode returns, particularly as strategy adoption increases. Second, the economic mechanisms underlying the ACU anomaly remain incompletely understood, and future research should investigate whether the premium reflects compensation for systematic risk or market inefficiencies. Third, our sample period and market coverage may limit the generalizability of findings, and out-of-sample validation across different market regimes and international markets would strengthen the evidence.

Future research should pursue several promising avenues. Investigating the time-varying nature of the ACU premium and its relationship to macroeconomic conditions could provide insights into the underlying economic drivers. Additionally, examining the interaction between ACU and corporate governance characteristics might reveal whether the anomaly is stronger among firms with particular organizational structures or managerial incentives. Finally, developing a theoretical framework that explains why acquisition capacity utilization predicts returns would advance our understanding of both asset pricing and corporate finance. Such work could explore

whether the ACU effect reflects rational pricing of acquisition-related risks or behavioral biases in how investors process complex corporate actions. Understanding these mechanisms is crucial for determining whether the ACU anomaly is likely to persist or diminish as markets evolve and become more efficient.

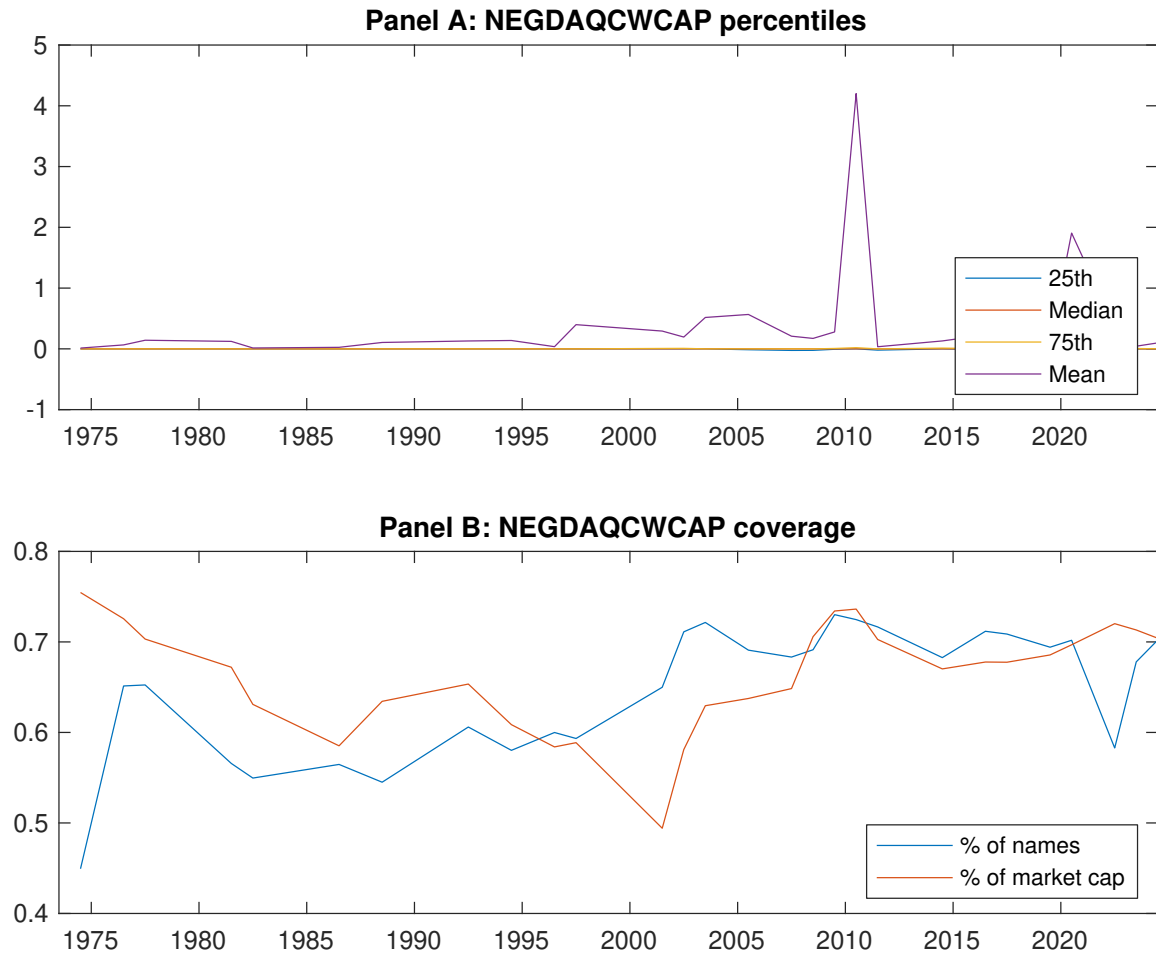


Figure 1: Times series of ACU percentiles and coverage.
This figure plots descriptive statistics for ACU. Panel A shows cross-sectional percentiles of ACU over the sample. Panel B plots the monthly coverage of ACU relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ACU. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on ACU-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [2.80]	0.74 [3.83]	0.67 [3.50]	0.63 [3.19]	0.75 [3.98]	0.21 [3.06]
α_{CAPM}	-0.11 [-1.88]	0.09 [1.58]	0.03 [0.48]	-0.03 [-0.50]	0.11 [2.14]	0.22 [3.21]
α_{FF3}	-0.10 [-1.66]	0.12 [2.07]	0.05 [0.78]	0.01 [0.12]	0.11 [2.07]	0.21 [2.98]
α_{FF4}	-0.08 [-1.43]	0.11 [1.93]	0.01 [0.11]	0.02 [0.25]	0.12 [2.12]	0.20 [2.84]
α_{FF5}	-0.19 [-3.39]	0.12 [2.07]	0.04 [0.71]	-0.04 [-0.66]	0.01 [0.10]	0.20 [2.75]
α_{FF6}	-0.18 [-3.08]	0.11 [1.97]	0.01 [0.21]	-0.03 [-0.44]	0.02 [0.32]	0.19 [2.66]
Panel B: Fama and French (2018) 6-factor model loadings for ACU-sorted portfolios						
β_{MKT}	1.01 [76.41]	0.98 [72.68]	0.96 [69.30]	1.02 [69.91]	0.99 [81.75]	-0.02 [-1.11]
β_{SMB}	0.03 [1.53]	0.02 [1.00]	0.08 [4.00]	-0.01 [-0.44]	0.08 [4.48]	0.05 [2.03]
β_{HML}	-0.10 [-3.84]	-0.09 [-3.31]	-0.07 [-2.63]	-0.19 [-6.84]	-0.08 [-3.33]	0.02 [0.62]
β_{RMW}	0.21 [8.12]	-0.03 [-0.96]	-0.04 [-1.29]	0.02 [0.76]	0.23 [9.54]	0.02 [0.49]
β_{CMA}	0.10 [2.64]	0.02 [0.53]	0.04 [1.00]	0.19 [4.44]	0.12 [3.43]	0.02 [0.40]
β_{UMD}	-0.03 [-1.95]	0.01 [0.55]	0.05 [3.49]	-0.02 [-1.44]	-0.02 [-1.56]	0.01 [0.41]
Panel C: Average number of firms (n) and market capitalization (me)						
n	453	654	783	672	459	
me (\$10 ⁶)	1663	1891	1211	1778	1702	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ACU strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21 [3.06]	0.22 [3.21]	0.21 [2.98]	0.20 [2.84]	0.20 [2.75]	0.19 [2.66]
Quintile	NYSE	EW	0.18 [4.05]	0.18 [4.13]	0.17 [3.83]	0.16 [3.50]	0.16 [3.55]	0.15 [3.37]
Quintile	Name	VW	0.13 [2.06]	0.14 [2.14]	0.13 [2.00]	0.15 [2.24]	0.13 [1.94]	0.14 [2.14]
Quintile	Cap	VW	0.19 [2.62]	0.21 [3.00]	0.20 [2.74]	0.18 [2.51]	0.19 [2.56]	0.18 [2.41]
Decile	NYSE	VW	0.18 [2.27]	0.21 [2.52]	0.19 [2.25]	0.18 [2.12]	0.17 [1.95]	0.16 [1.88]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.17 [2.38]	0.17 [2.38]	0.15 [2.17]	0.15 [2.11]	0.13 [1.75]	0.13 [1.74]
Quintile	NYSE	EW	-0.03 [-0.60]					
Quintile	Name	VW	0.09 [1.35]	0.09 [1.38]	0.08 [1.23]	0.09 [1.39]	0.07 [0.98]	0.07 [1.12]
Quintile	Cap	VW	0.14 [1.97]	0.16 [2.21]	0.14 [1.95]	0.13 [1.84]	0.12 [1.65]	0.12 [1.59]
Decile	NYSE	VW	0.14 [1.65]	0.14 [1.66]	0.12 [1.41]	0.11 [1.35]	0.08 [0.95]	0.08 [0.95]

Table 3: Conditional sort on size and ACU

This table presents results for conditional double sorts on size and ACU. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ACU. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ACU and short stocks with low ACU. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ACU Quintiles					ACU Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.79 [2.86]	0.84 [2.93]	0.81 [2.86]	0.91 [3.23]	0.90 [3.04]	0.11 [0.89]	0.10 [0.86]	0.10 [0.79]	0.04 [0.35]	0.04 [0.30]	0.00 [0.03]
	(2)	0.81 [3.15]	0.86 [3.32]	0.85 [3.19]	0.88 [3.38]	0.86 [3.34]	0.05 [0.62]	0.06 [0.77]	0.02 [0.23]	0.01 [0.08]	-0.01 [-0.13]	-0.02 [-0.23]
	(3)	0.79 [3.37]	0.82 [3.41]	0.90 [3.77]	0.79 [3.24]	0.87 [3.68]	0.08 [0.97]	0.08 [0.94]	0.05 [0.63]	0.04 [0.45]	0.06 [0.68]	0.05 [0.55]
	(4)	0.77 [3.53]	0.88 [3.78]	0.85 [3.91]	0.85 [3.72]	0.81 [3.70]	0.04 [0.52]	0.03 [0.43]	0.04 [0.49]	0.05 [0.63]	0.01 [0.14]	0.02 [0.31]
	(5)	0.46 [2.35]	0.71 [3.77]	0.60 [3.25]	0.57 [2.92]	0.76 [4.08]	0.30 [3.46]	0.32 [3.66]	0.30 [3.41]	0.28 [3.16]	0.29 [3.23]	0.28 [3.04]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ACU Quintiles					ACU Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	351	350	349	349	349	34	26	27	26	33	
	(2)	93	93	93	93	93	52	52	50	51	52	
	(3)	64	64	64	64	64	89	87	86	88	90	
	(4)	52	53	53	53	52	193	197	191	194	195	
(5)	45	45	45	45	45	1176	1494	1070	1437	1254		

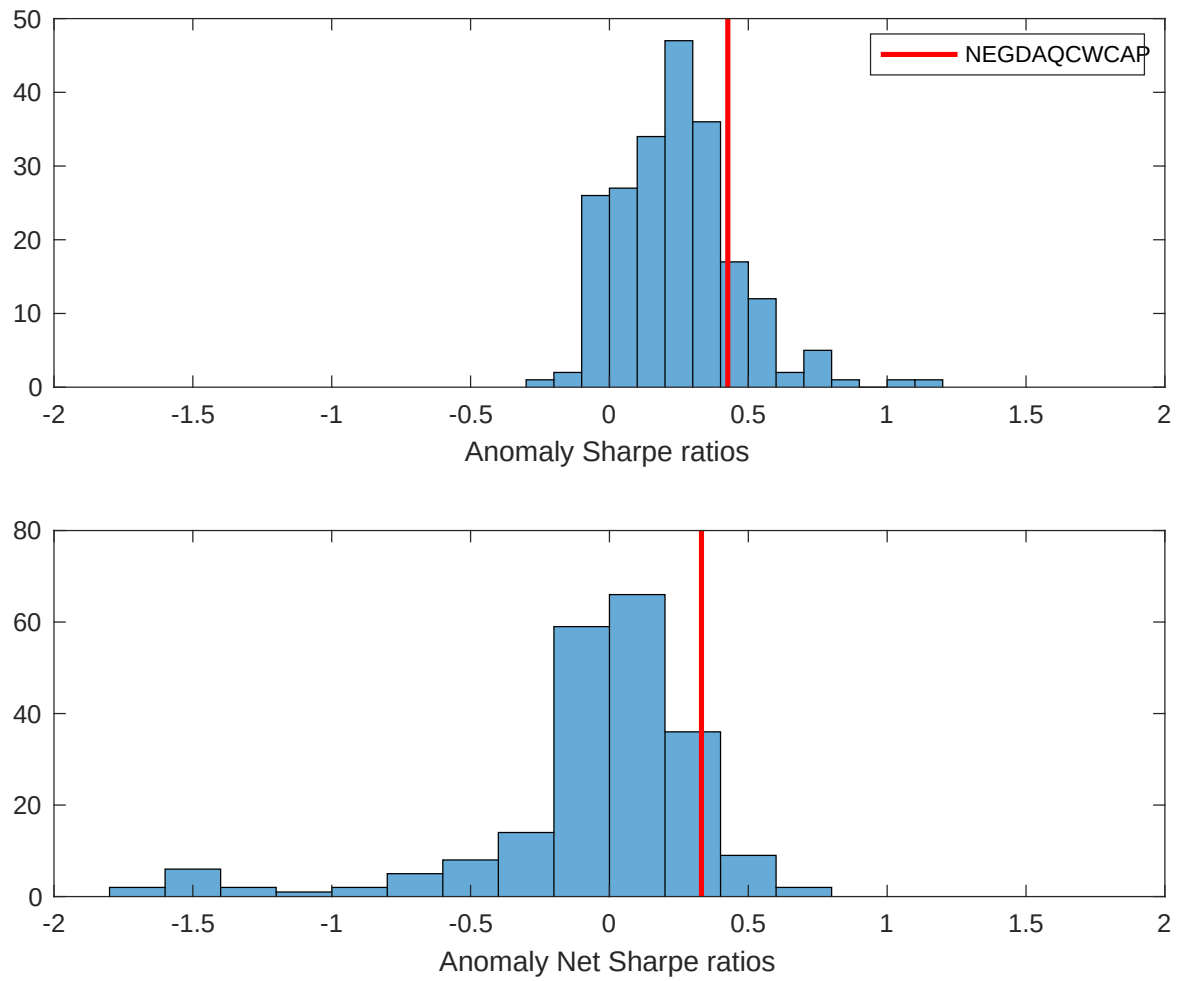


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ACU with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

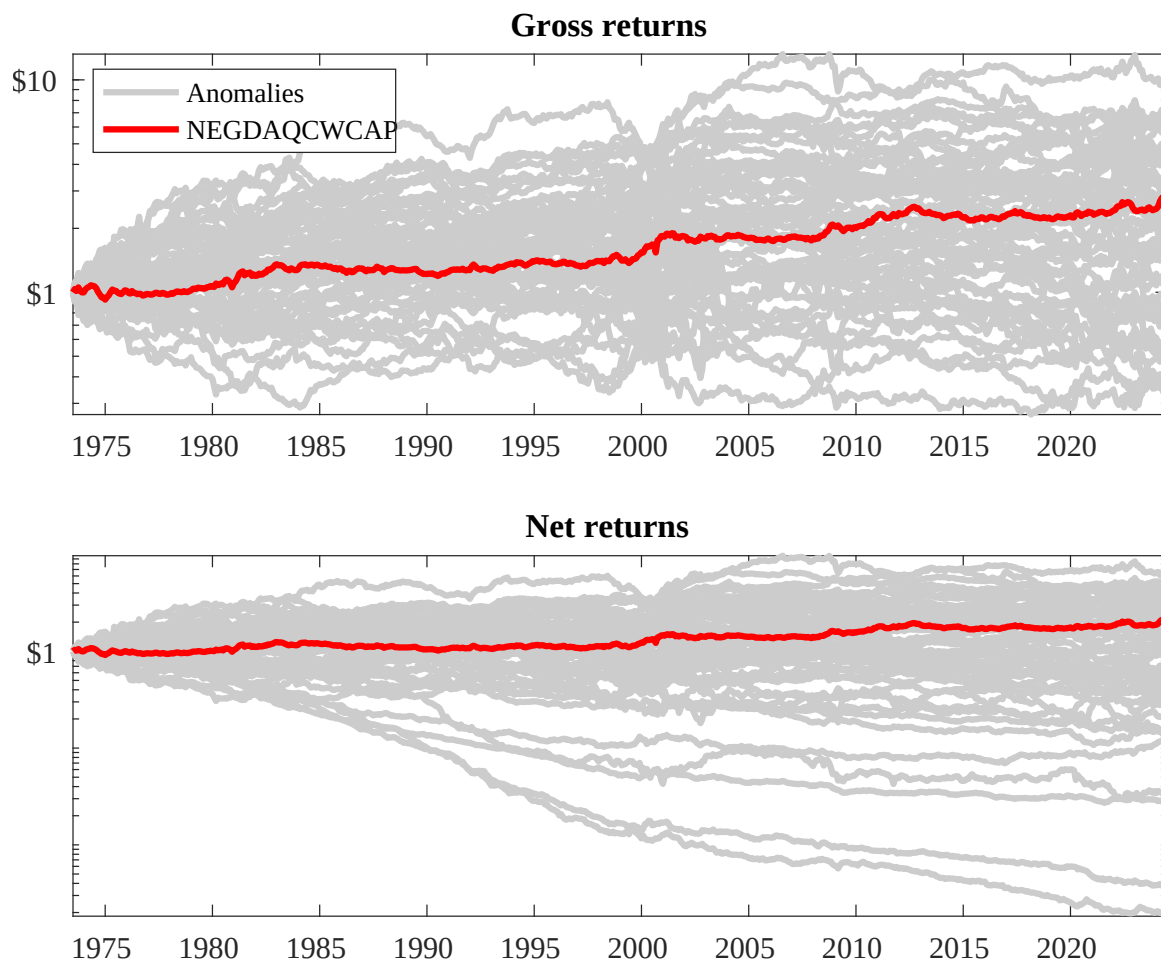
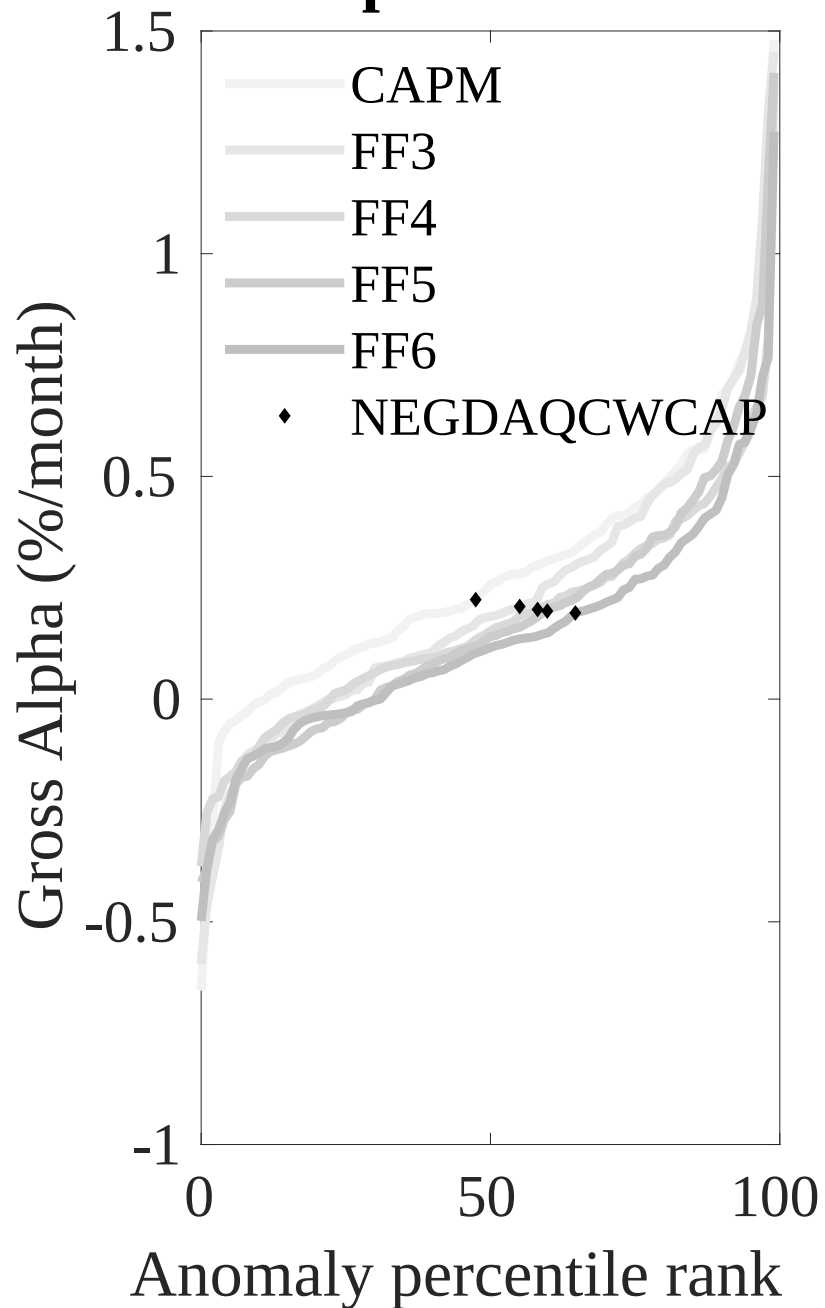


Figure 3: Dollar invested.

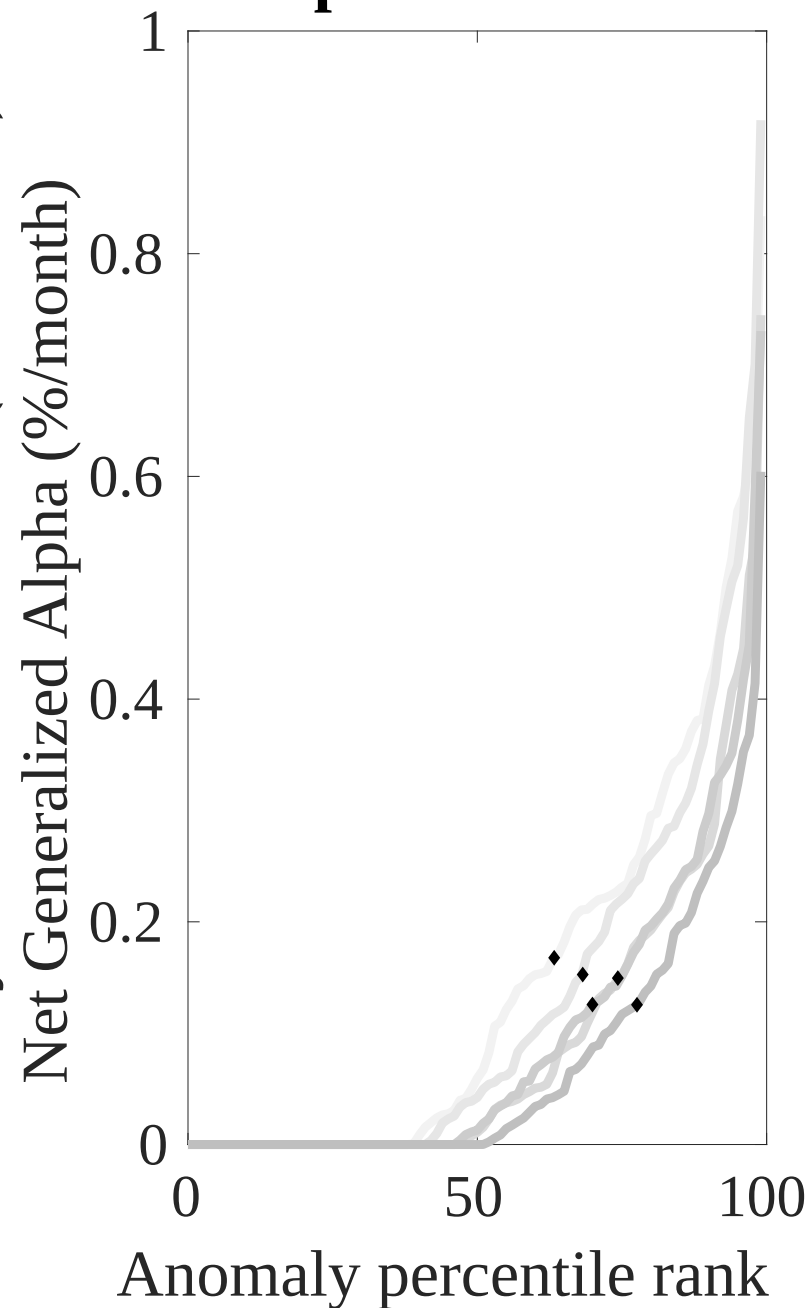
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ACU trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



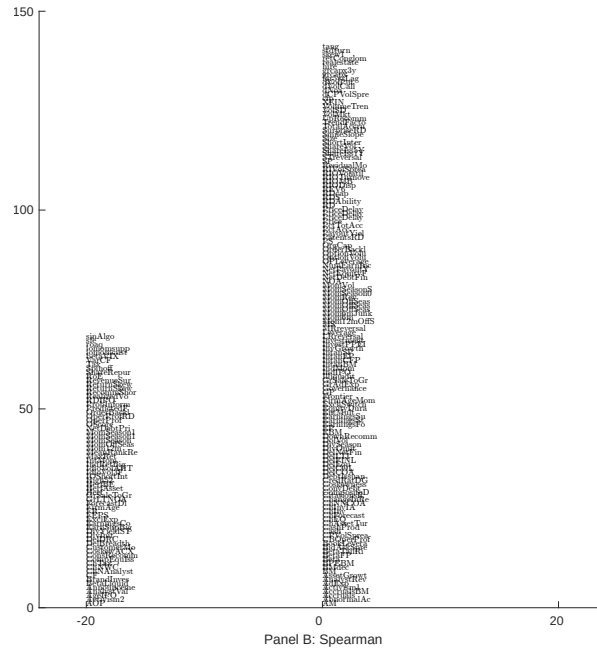
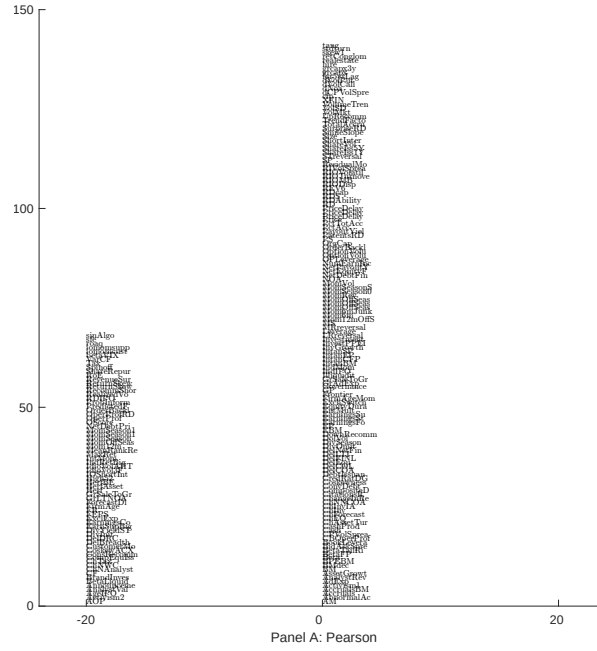


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with ACU. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

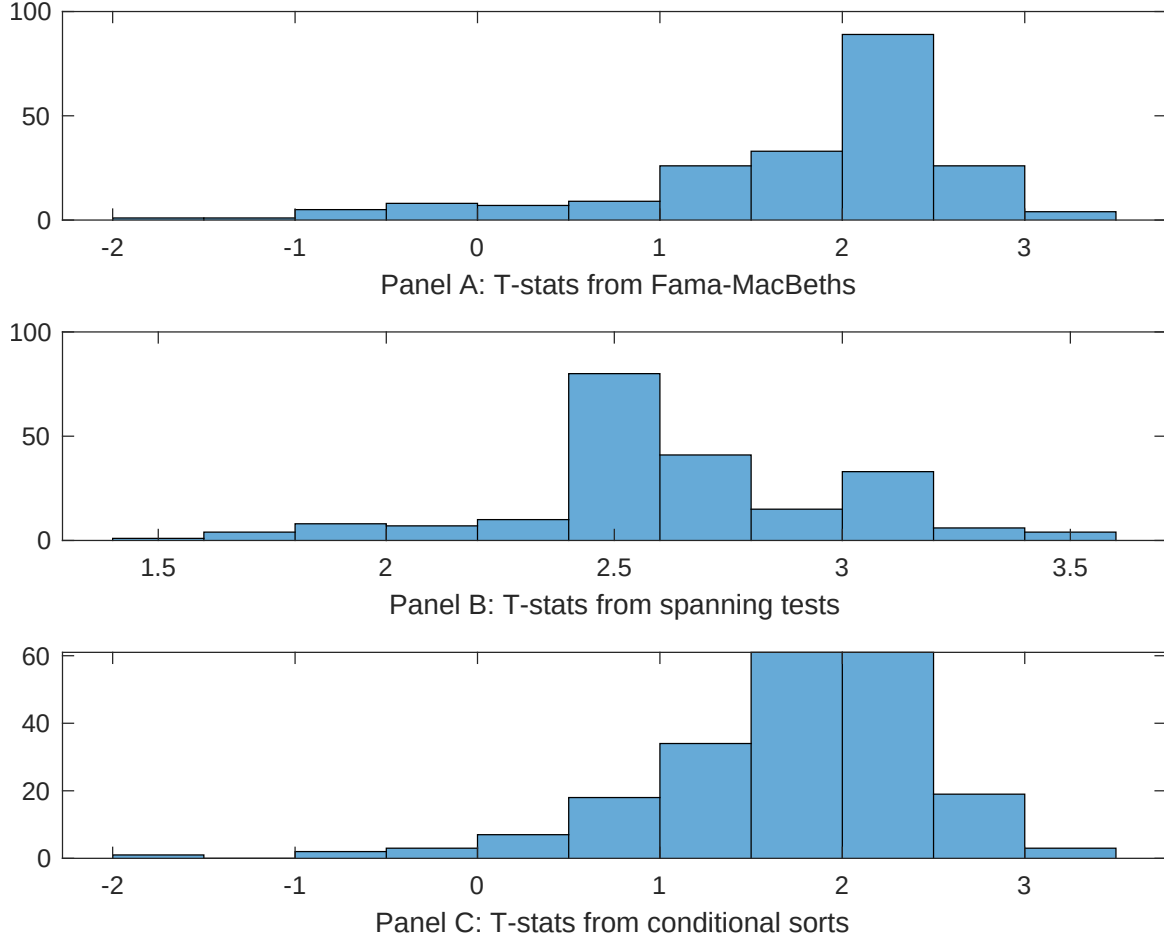


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ACU conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ACU} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ACU}ACU_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ACU,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ACU. Stocks are finally grouped into five ACU portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ACU trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ACU. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ACU}ACU_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in net financial assets, Change in Net Noncurrent Op Assets, Asset growth, Growth in book equity, Change in equity to assets, Sales growth over inventory growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.18]	0.13 [5.09]	0.14 [5.72]	0.19 [6.95]	0.14 [5.30]	0.13 [5.18]	0.15 [6.81]
ACU	0.22 [0.89]	0.47 [1.90]	-0.17 [-0.65]	0.52 [2.14]	0.49 [2.01]	0.53 [2.13]	-0.23 [-0.88]
Anomaly 1	0.79 [5.24]						0.43 [1.76]
Anomaly 2		0.92 [4.25]					0.99 [0.31]
Anomaly 3			0.11 [8.51]				0.84 [6.06]
Anomaly 4				0.52 [5.13]			0.33 [0.30]
Anomaly 5					0.15 [4.35]		0.49 [0.79]
Anomaly 6						0.13 [3.58]	0.60 [1.56]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ACU trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ACU} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in net financial assets, Change in Net Noncurrent Op Assets, Asset growth, Growth in book equity, Change in equity to assets, Sales growth over inventory growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.21 [2.93]	0.19 [2.62]	0.18 [2.44]	0.17 [2.37]	0.18 [2.45]	0.18 [2.54]	0.19 [2.71]
Anomaly 1	-15.30 [-4.50]						-5.87 [-1.46]
Anomaly 2		-15.54 [-4.81]					-10.57 [-3.03]
Anomaly 3			13.88 [3.31]				3.86 [0.81]
Anomaly 4				17.06 [4.42]			8.50 [1.51]
Anomaly 5					14.49 [4.08]		1.86 [0.35]
Anomaly 6						-10.73 [-3.45]	-5.38 [-1.68]
mkt	-2.20 [-1.33]	-1.95 [-1.18]	-1.88 [-1.13]	-1.30 [-0.79]	-1.99 [-1.21]	-1.68 [-1.01]	-1.65 [-1.01]
smb	4.11 [1.64]	5.80 [2.32]	3.99 [1.57]	4.47 [1.79]	5.05 [2.02]	5.45 [2.17]	4.89 [1.95]
hml	1.89 [0.59]	2.22 [0.70]	0.90 [0.28]	-0.42 [-0.13]	-0.40 [-0.12]	1.26 [0.39]	0.52 [0.16]
rmw	0.28 [0.08]	4.07 [1.25]	2.16 [0.66]	2.67 [0.82]	3.41 [1.05]	3.25 [1.00]	3.77 [1.14]
cma	-0.99 [-0.20]	2.49 [0.52]	-13.81 [-1.97]	-13.35 [-2.20]	-11.46 [-1.91]	3.66 [0.75]	-13.13 [-1.83]
umd	0.84 [0.51]	2.64 [1.56]	0.84 [0.51]	0.12 [0.07]	0.67 [0.41]	1.82 [1.08]	2.57 [1.51]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	3	4	2	3	3	2	7

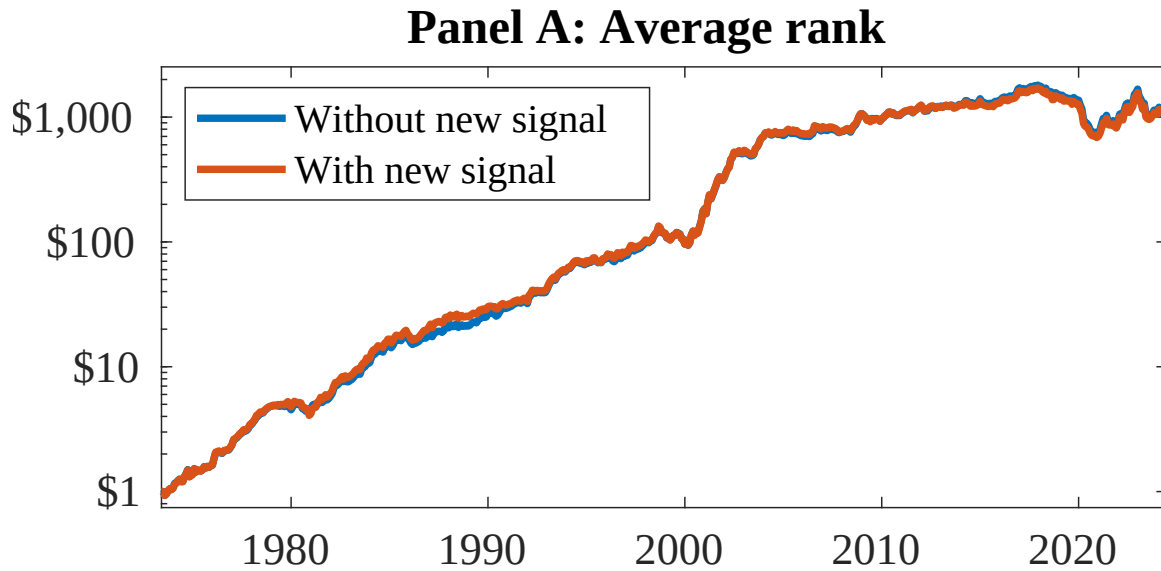


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as ACU. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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